
FORTIGATE NSE4

Lab 3 — High Availability Active-Passive Configuration

A complete lab deploying two FortiGate firewalls in Active-Passive HA — covering topology planning with dedicated heartbeat links, interface and policy pre-configuration, HA cluster setup via GUI, FGCP protocol fundamentals, priority-based primary election, failover testing (power-off and WAN disconnect), and continuous ping verification.

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Category: FortiGate NSE4 Lab Series · Portfolio Project

Platform: 2x FortiGate VM v7.6.5 on EVE-NG · Active-Passive HA (FGCP)

1. Project Goal

Deploy two FortiGate firewalls in High Availability (Active-Passive) mode to provide zero/minimal downtime, session synchronization, and automatic failover for enterprise and OT environments.

► 1.1 Why Active-Passive?

- Most common HA mode in enterprise and OT environments
- Predictable behavior — only one unit processes traffic at a time
- Easier troubleshooting compared to Active-Active
- Required for many OT environments where deterministic failover is critical

► 1.2 Key Rules

- Heartbeat interfaces must be directly connected (dedicated cable/link)
- Both FortiGates must run the same firmware version
- Both must have identical interface configurations
- The same group name and password must be configured on both units

2. Lab Topology

The topology uses two FortiGate VMs, each with four ports. A dedicated heartbeat link (port4) connects the two firewalls directly for HA synchronization:

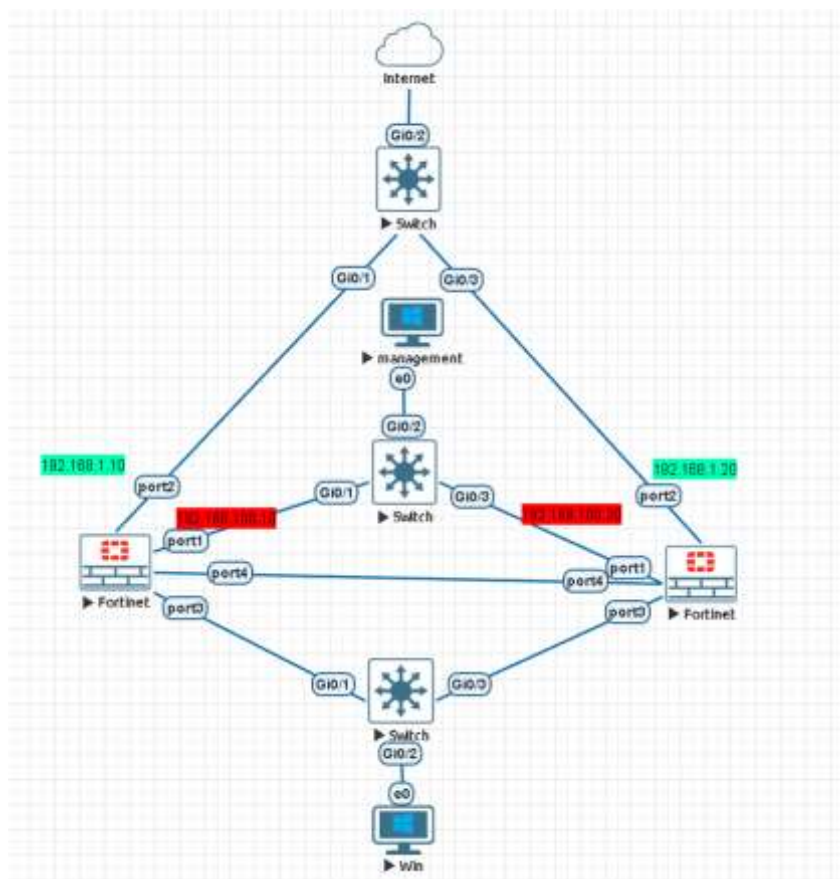


Figure 1 — HA Topology: Two FortiGates with shared LAN/WAN and dedicated heartbeat (port4)

Port	FortiGate 1 (Primary)	FortiGate 2 (Secondary)
port1 (MGMT)	192.168.100.10/24	192.168.100.20/24
port2 (WAN)	192.168.1.9/24 (internet)	192.168.1.20/24
port3 (LAN)	192.168.20.1/24	192.168.20.1/24 (synced)
port4 (Heartbeat)	Direct link to peer	Direct link to peer

3. Pre-HA Configuration

Before enabling HA, configure the primary FortiGate with all necessary interfaces, routing, and policies. The secondary will receive this configuration automatically through HA synchronization.

► 3.1 Verify WAN Connectivity

Test internet reachability from the FortiGate CLI:

```
FortiGate-VM64-KVM # execute ping 8.8.8.8
```

```
FortiGate-VM64-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=120 time=50.0 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=120 time=47.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=120 time=48.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=120 time=48.8 ms
```

Figure 2 — WAN connectivity verified: ping 8.8.8.8 successful

► 3.2 Routing Table

Verify the routing table shows the default route and connected interfaces:

```
FortiGate-VM64-KVM # get router info routing-table all
```

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
V - BGP VPNv4
* - candidate default

Routing table for VRF=0
S* 0.0.0.0/0 [5/0] via 192.168.1.1, port2, [1/0]
C 192.168.1.0/24 is directly connected, port2
C 192.168.20.0/24 is directly connected, port3
C 192.168.100.0/24 is directly connected, port1
```

Figure 3 — Routing table: default route via port2, connected subnets

► 3.3 Interface Summary

Physical Interface			
LAN (port3)	Physical Interface	192.168.20.1/255.255.255.0	
port1	Physical Interface	192.168.100.10/255.255.255.0	PING HTTPS SSH HTTP
port2	Physical Interface	192.168.1.9/255.255.255.0	
port4	Physical Interface	0.0.0.0/0.0.0.0	

Figure 4 — Interface list: port1 (MGMT), port2 (WAN), port3 (LAN), port4 (Heartbeat)

► 3.4 Firewall Policy

Create a LAN-to-WAN policy allowing internet access with NAT and DNS/HTTPS/PING services:



Figure 5 — Firewall policy: LAN (port3) → WAN (port2), ACCEPT, NAT, UTM

4. HA Configuration

► 4.1 Navigate to HA Settings

On each FortiGate, go to System → HA. The default mode is Standalone — change it to Active-Passive:

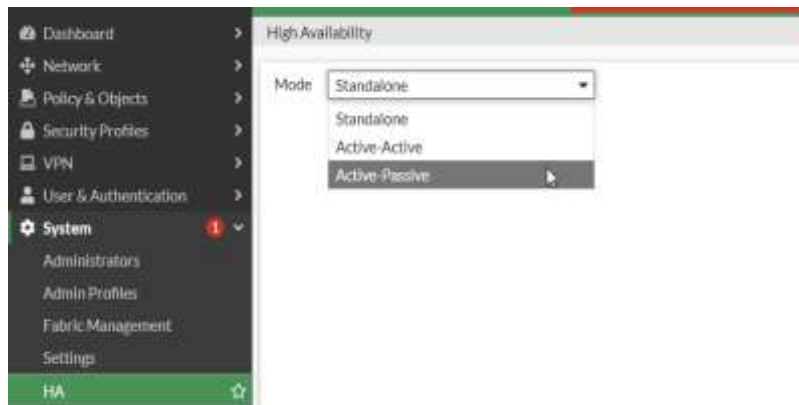


Figure 6 — System → HA: Mode selection (Standalone / Active-Active / Active-Passive)

► 4.2 Primary FortiGate Configuration

Configure the primary unit with a higher priority value (128) so it becomes the active unit:



Figure 7 — Primary HA config: priority 128, group HA-PAIR, heartbeat port4

Setting	Primary Value
Mode	Active-Passive
Device Priority	128 (higher = primary)
Group Name	HA-PAIR
Password	(shared secret)
Session Pickup	Disabled (on primary)
Monitor Interfaces	port2 (WAN), LAN (port3)
Heartbeat Interfaces	port4

► 4.3 Secondary FortiGate Configuration

Configure the secondary unit with a lower priority (100) and enable Session Pickup so existing connections survive failover:

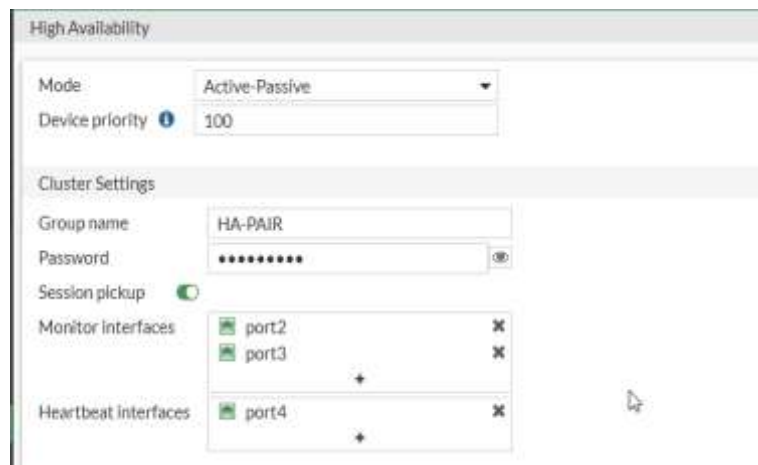


Figure 8 — Secondary HA config: priority 100, session pickup enabled

Setting	Secondary Value
Mode	Active-Passive
Device Priority	100 (lower = secondary)
Group Name	HA-PAIR (must match)
Password	(must match primary)
Session Pickup	Enabled
Monitor Interfaces	port2, port3
Heartbeat Interfaces	port4

► 4.4 Priority & Primary Election

FortiGate uses HA priority to determine which unit becomes the Primary (Active):

- Higher priority = higher chance to become Primary
- Priority range: 0–255
- Default priority: 128
- If priorities are equal, the unit with the higher uptime wins

- If both are equal, the unit with the higher serial number wins

After applying HA settings, the secondary device will lose independent management access — it is now controlled exclusively through the primary.

5. Verification

► 5.1 HA Status

After HA formation, the primary FortiGate dashboard shows both units synchronized:

Status	Priority	Hostname	Serial No.	Role	System Uptime	Sessions	Throughput
Synchronized	128	FortiGate1	FGVMEV1405X04008	Primary	3m 46s	27	19.00 kbps
Synchronized	120	FortiGate-VM6	FGVMEV2F7FDQJ0D5	Secondary	3m 48s	4	16.00 kbps

Figure 9 — HA Status: FortiGate1 (Primary, priority 128) + FortiGate-VM6 (Secondary, priority 120)

Key indicators: both units show "Synchronized" status, the primary is handling 27 sessions at 19 kbps, and both have matching uptime.

► 5.2 Connectivity Test

From the test PC, run a continuous ping to verify internet access through the HA pair:

```
C:\Users\user> ping 8.8.8.8 -t
```

```
C:\Users\user>ping 8.8.8.8 -t

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=52ms TTL=118
Reply from 8.8.8.8: bytes=32 time=53ms TTL=118
Reply from 8.8.8.8: bytes=32 time=52ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=53ms TTL=118
```

Figure 10 — Continuous ping to 8.8.8.8: all replies successful

6. Failover Testing

Failover testing is critical — you MUST test this in every HA deployment to validate that the secondary takes over correctly.

► 6.1 Test Scenarios

1. Power off the primary FortiGate
2. Disconnect the WAN port (port2) from the primary
3. Disconnect the heartbeat cable (port4)

► 6.2 Expected Results

- Secondary becomes primary within 1–5 seconds
- Continuous ping drops 1–2 packets maximum
- Existing sessions survive if session pickup is enabled

► 6.3 WAN Link Removal Test

The WAN link was disconnected from the primary FortiGate to simulate an upstream failure:

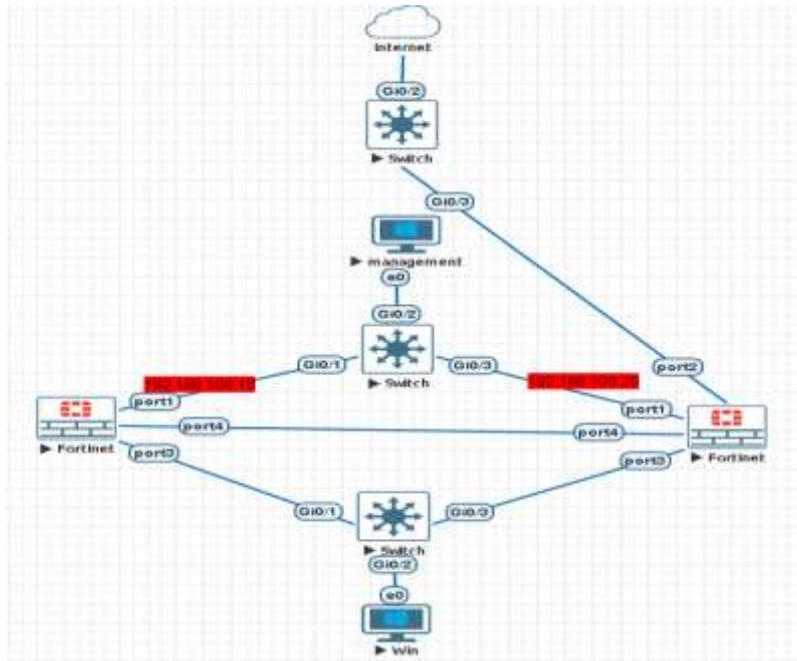


Figure 11 — Topology during failover test: primary WAN link removed

The continuous ping continued without interruption, confirming the secondary assumed the active role and maintained connectivity:

```

Reply from 8.8.8.8: bytes=32 time=50ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=52ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=55ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=52ms TTL=118
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Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=50ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118

```

Figure 12 — Continuous ping during WAN failover: zero packet loss

The HA pair maintained connectivity throughout the failover — the ping stream shows no gaps or timeouts, confirming seamless transition.

7. FGCP — FortiGate Clustering Protocol

FGCP (FortiGate Clustering Protocol) is the proprietary protocol FortiGate uses to build and maintain High Availability clusters.

► 7.1 What FGCP Does

- Heartbeat monitoring — detects peer failures via dedicated heartbeat links
- Configuration synchronization — replicates config from primary to secondary
- Session synchronization — replicates active sessions (when session pickup is enabled)
- Primary election — determines which unit becomes active based on priority, uptime, and serial number
- Virtual MAC — the HA pair shares a virtual MAC address so failover is transparent to the network

► 7.2 FGCP vs. Cisco FGCP Equivalent

Feature	FortiGate (FGCP)	Cisco FTD (Failover)
Protocol	FGCP (proprietary)	Cisco failover protocol
Heartbeat	Dedicated heartbeat interface(s)	Failover + State link
Session Sync	Session pickup (toggle)	Stateful failover (always)
Primary Election	Priority (higher = primary)	Primary/Secondary role assignment
Config Sync	Automatic from primary	Automatic from active
Virtual MAC	Yes — FGCP vMAC	Yes — vMAC

8. Summary

Component	Details
HA Mode	Active-Passive
Protocol	FGCP (FortiGate Clustering Protocol)
Primary Election	Higher priority wins (range 0–255, default 128)
Heartbeat Interface	port4 (dedicated, directly connected)
Monitor Interfaces	port2 (WAN) + port3 (LAN)
Session Pickup	Enabled on secondary for session survivability
Config Sync	Automatic from primary to secondary
Failover Time	1–5 seconds (1–2 packets lost maximum)
Failover Tests	Power off, WAN disconnect, heartbeat disconnect
Verification	HA Status dashboard + continuous ping
OT Relevance	Deterministic failover required for industrial process continuity

